

Sheth L.U.J. And Sir M.V. College

Aim:- Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

CLI VERSION:-

```
Practical No. 3 Cyber Security.py - C:/Users/mvlul/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help
import hmac
import hashlib

print("=" * 55)
print("MESSAGE AUTHENTICATION CODE (HMAC - SHA256)")
print("=" * 55)

secret_key = input("Enter Secret Key: ").encode()

message = input("Enter Message: ").encode()

mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:")
print(mac)

print("\n----- Verification -----")

received_message = input("Enter Message to Verify: ").encode()
received_mac = input("Enter Received MAC: ")

new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()
```

```
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secret_key = input("Enter Secret Key: ").encode()

message = input("Enter Message: ").encode()

mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:")
print(mac)

print("\n----- Verification -----")

received_message = input("Enter Message to Verify: ").encode()
received_mac = input("Enter Received MAC: ")

new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

if hmac.compare_digest(new_mac, received_mac):
    print("\n✅ MAC Verified Successfully!")
    print("Data Integrity Maintained.")
    print("Message is Authentic.")
else:
    print("\n❌ MAC Verification Failed!")
    print("Message has been Modified or Key is Incorrect.")
```

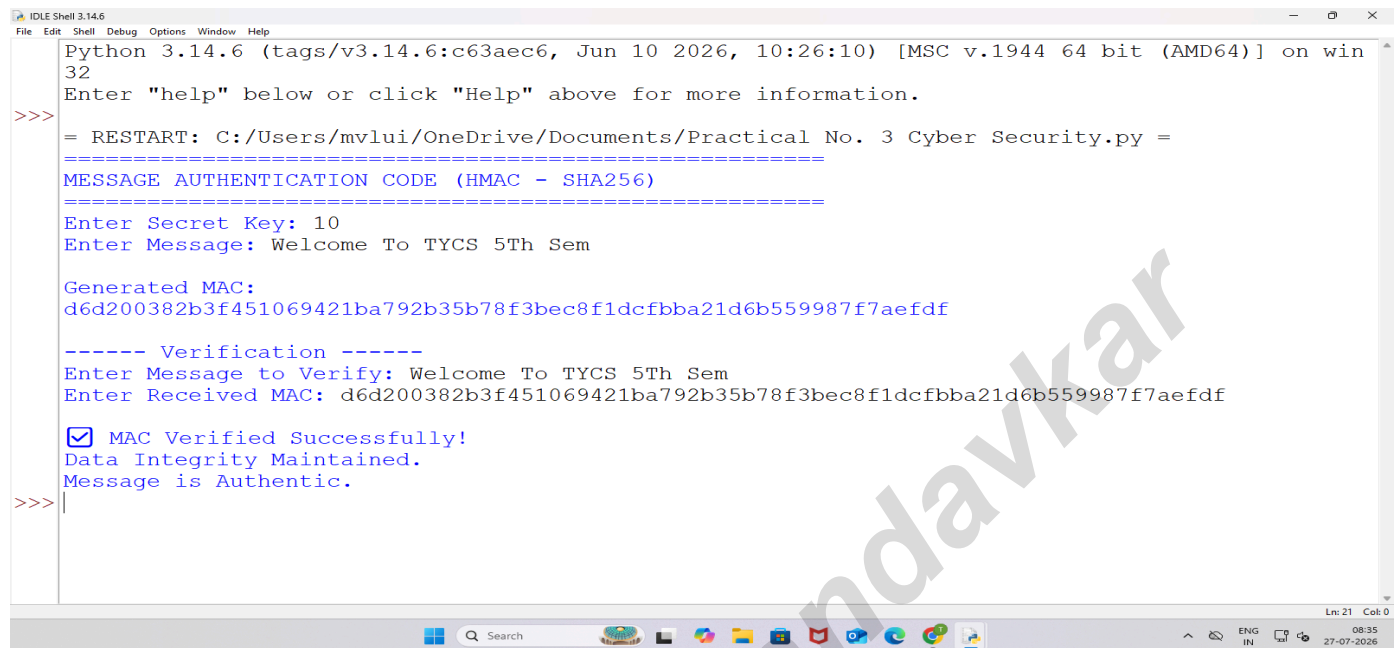
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Cyber And Information Security

Sheth L.U.J. And Sir M.V. College

OUTPUT:-



```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win
32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/mvlui/OneDrive/Documents/Practical No. 3 Cyber Security.py =
=====
MESSAGE AUTHENTICATION CODE (HMAC - SHA256)
=====
Enter Secret Key: 10
Enter Message: Welcome To TYCS 5Th Sem

Generated MAC:
d6d200382b3f451069421ba792b35b78f3bec8f1dcfbba21d6b559987f7aefdf

----- Verification -----
Enter Message to Verify: Welcome To TYCS 5Th Sem
Enter Received MAC: d6d200382b3f451069421ba792b35b78f3bec8f1dcfbba21d6b559987f7aefdf

☒ MAC Verified Successfully!
Data Integrity Maintained.
Message is Authentic.
>>> |
```

GUI VERSION:-



```
Practical No. 3 Cyber Security.py - C:/Users/mvlui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

def generate_mac():
    key = entry_key.get().strip().encode()
    msg = txt_message.get("1.0", tk.END).strip().encode()

    if not key or not msg:
        messagebox.showerror("Error", "Please enter Secret Key and Message.")
        return

    mac = hmac.new(key, msg, hashlib.sha256).hexdigest()

    txt_mac.config(state="normal")
    txt_mac.delete("1.0", tk.END)
    txt_mac.insert(tk.END, mac)
    txt_mac.config(state="disabled")

def verify_mac():
    key = entry_key.get().strip().encode()
    msg = txt_verify_message.get("1.0", tk.END).strip().encode()
```

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```
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def verify_mac():
    key = entry_key.get().strip().encode()
    msg = txt_verify_message.get("1.0", tk.END).strip().encode()
    received_mac = txt_received_mac.get("1.0", tk.END).strip()

    if not key or not msg or not received_mac:
        messagebox.showerror("Error", "Please fill all fields.")
        return

    new_mac = hmac.new(key, msg, hashlib.sha256).hexdigest()

    if hmac.compare_digest(new_mac, received_mac):
        lbl_result.config(
            text="✔ MAC VERIFIED SUCCESSFULLY",
            fg="#00ff66"
        )
        messagebox.showinfo("Success",
            "Message is Authentic.\nData Integrity Maintained.")
    else:
        lbl_result.config(
            text="✘ MAC VERIFICATION FAILED",
            fg="red"
        )
        messagebox.showerror("Failed",
            "Message has been Modified\nor Secret Key is Incorrect.")

Ln: 325 Col: 0
08:53 27-07-2026
```

```
Practical No. 3 Cyber Security.py - C:/Users/mvlui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

def copy_mac():
    txt = txt_mac.get("1.0", tk.END).strip()

    if txt == "":
        return

    root.clipboard_clear()
    root.clipboard_append(txt)

    messagebox.showinfo("Copied", "Generated MAC copied to Clipboard.")

def clear_all():
    entry_key.delete(0, tk.END)

    txt_message.delete("1.0", tk.END)
    txt_mac.config(state="normal")
    txt_mac.delete("1.0", tk.END)
    txt_mac.config(state="disabled")

    txt_verify_message.delete("1.0", tk.END)
    txt_received_mac.delete("1.0", tk.END)

    lbl_result.config(text="")

Ln: 325 Col: 0
08:53 27-07-2026
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```
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File Edit Format Run Options Window Help

root = tk.Tk()

root.title("Cyber Security - Message Authentication Code")
root.geometry("900x760")
root.configure(bg="#0b0f19")
root.resizable(False, False)

title = tk.Label(
    root,
    text="🔐 MESSAGE AUTHENTICATION CODE (HMAC - SHA256)",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Consolas", 20, "bold")
)

title.pack(pady=20)

subtitle = tk.Label(
    root,
    text="Cyber & Information Security Practical",
    bg="#0b0f19",
    fg="white",
    font=("Calibri", 12)
)

Ln: 325 Col: 0
08:54
27-07-2026
```

```
Practical No. 3 Cyber Security.py - C:/Users/mlvui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

subtitle.pack()

tk.Label(
    root,
    text="Secret Key",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Arial", 11, "bold")
).pack(anchor="w", padx=40)

entry_key = tk.Entry(
    root,
    width=80,
    font=("Consolas", 11),
    bg="#1b263b",
    fg="white",
    insertbackground="white"
)

entry_key.pack(padx=40, pady=5)

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tk.Label(
    root,
    text="Original Message",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Arial", 11, "bold")
).pack(anchor="w", padx=40)

txt_message = tk.Text(
    root,
    height=5,
    width=95,
    bg="#1b263b",
    fg="white",
    font=("Consolas", 10),
    insertbackground="white"
)

txt_message.pack(padx=40)

tk.Button(
    root,
    text="Generate MAC",
    bg="#008cff",
    command=generate_mac
).pack(pady=15)
```

```
Practical No. 3 Cyber Security.py - C:/Users/mvlui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

tk.Button(
    root,
    text="Generate MAC",
    bg="#008cff",
    fg="white",
    font=("Arial", 11, "bold"),
    width=20,
    command=generate_mac
).pack(pady=15)

tk.Label(
    root,
    text="Generated MAC",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Arial", 11, "bold")
).pack(anchor="w", padx=40)

frame_mac = tk.Frame(root)
frame_mac.pack()

scroll = tk.Scrollbar(frame_mac)
```

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```
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File Edit Format Run Options Window Help

scroll = tk.Scrollbar(frame_mac)

scroll.pack(side=tk.RIGHT, fill=tk.Y)

txt_mac = tk.Text(
    frame_mac,
    height=4,
    width=95,
    bg="black",
    fg="#00ff66",
    font=("Consolas", 10),
    yscrollcommand=scroll.set,
    state="disabled"
)

txt_mac.pack(side=tk.LEFT)

scroll.config(command=txt_mac.yview)

tk.Button(
    root,
    text="Copy Generated MAC",
    bg="#00aa88",
    fg="white",
    width=22,
    font=("Arial", 10, "bold"),
    command=copy_mac
).pack(pady=10)
```

```
Practical No. 3 Cyber Security.py - C:/Users/mvlui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

tk.Button(
    root,
    text="Copy Generated MAC",
    bg="#00aa88",
    fg="white",
    width=22,
    font=("Arial", 10, "bold"),
    command=copy_mac
).pack(pady=10)

tk.Label(
    root,
    text="Message for Verification",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Arial", 11, "bold")
).pack(anchor="w", padx=40)

txt_verify_message = tk.Text(
    root,
    height=5,
    width=95,
    bg="#1b263b",
    fg="white",
    font=("Consolas", 10)
)
```

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txt_verify_message = tk.Text(
    root,
    height=5,
    width=95,
    bg="#1b263b",
    fg="white",
    font=("Consolas", 10),
    insertbackground="white"
)

txt_verify_message.pack(padx=40)

tk.Label(
    root,
    text="Received MAC",
    bg="#0b0f19",
    fg="#00ff99",
    font=("Arial", 11, "bold")
).pack(anchor="w", padx=40)

txt_received_mac = tk.Text(
    root,
    height=4,
    width=95,
    bg="#1b263b",
    fg="white",
    font=("Consolas", 10),
    insertbackground="white"
)

txt_received_mac.pack(padx=40)

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27-07-2026
```

```
Practical No. 3 Cyber Security.py - C:/Users/mvui/OneDrive/Documents/Practical No. 3 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

txt_received_mac = tk.Text(
    root,
    height=4,
    width=95,
    bg="#1b263b",
    fg="white",
    font=("Consolas", 10),
    insertbackground="white"
)

txt_received_mac.pack(padx=40)

frame = tk.Frame(root, bg="#0b0f19")
frame.pack(pady=15)

tk.Button(
    frame,
    text="Verify MAC",
    bg="#00cc66",
    fg="white",
    width=18,
    font=("Arial", 11, "bold"),
    command=verify_mac
).grid(row=0, column=0, padx=15)

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File Edit Format Run Options Window Help

tk.Button(
    frame,
    text="Verify MAC",
    bg="#00cc66",
    fg="white",
    width=18,
    font=("Arial", 11, "bold"),
    command=verify_mac
).grid(row=0, column=0, padx=15)

tk.Button(
    frame,
    text="Clear",
    bg="#ff4444",
    fg="white",
    width=18,
    font=("Arial", 11, "bold"),
    command=clear_all
).grid(row=0, column=1, padx=15)

lbl_result = tk.Label(
    root,
    text="",
    bg="#0b0f19",
    fg="white",
    font=("Arial", 16, "bold")
)

lbl_result.pack(pady=15)

footer = tk.Label(
    root,
    text="Developed for Cyber & Information Security Practical",
    bg="#0b0f19",
    fg="gray",
    font=("Calibri", 10)
)

footer.pack(side="bottom", pady=10)

root.mainloop()
```

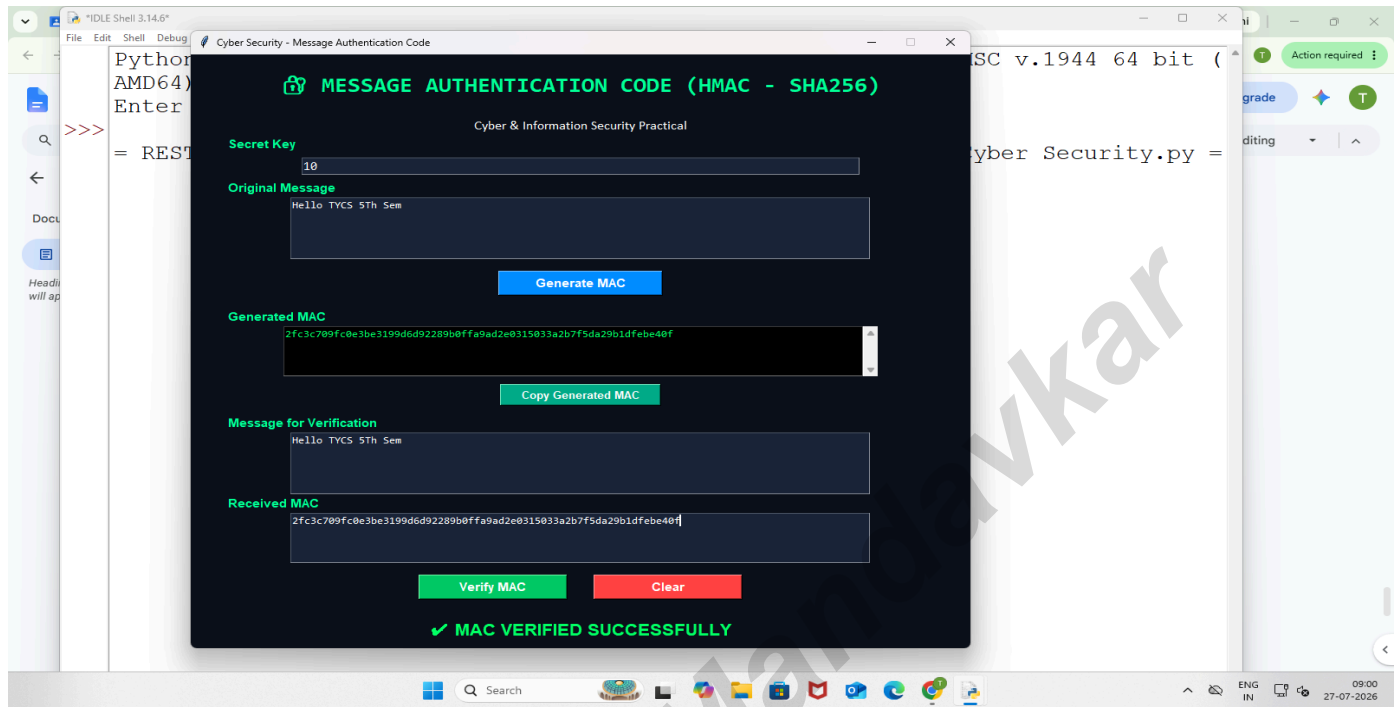
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OUTPUT:-



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